

Alexander Belik

Dublin, Ireland

Phone: +353 85 256 4027

✉ abelik1@outlook.com

🔗 linkedin.com/in/alexander-belik

🔗 github.com/Abelik1

Education

2022–2026. B.A. Theoretical Physics (expected)

Trinity College Dublin,
Dublin, Ireland

Focus: quantum mechanics,
statistical mechanics,
quantum field theory,
computational physics.

2020–2022. Leaving Certificate

The Dublin Academy of
Education, Dublin, Ireland

Skills

- Python (NumPy, automation, tooling)
- PyTorch / machine learning
- C/C++ computer simulations
- Scientific computing & data analysis (NumPy, Matplotlib, Mathematica)
- LaTeX (technical writing)
- Git / GitHub / Copilot
- UX collaboration (Figma) · Cursor / Claude

Languages

English — Fluent

Russian — Intermediate

Spanish — Basic

Interests

Programming projects, AI, travel, snowboarding, and speed-solving Rubiks Cubes (sub-25s).

About Me

Fourth-year Theoretical Physics student at Trinity College Dublin with a strong focus on software development, computational problem-solving, and applied machine learning. Experienced in building reliable Python tooling, simulation code, and data pipelines in both research and production environments. Comfortable translating complex requirements into practical, maintainable software through collaboration with technical and non-technical teams. I also have practical experience using modern AI tools, including large language models and lightweight agentic workflows, to support software development, analysis, and experimentation in a reliable and disciplined way.

Work Experience

Oct 2025 – Present Teaching Assistant

Trinity College Dublin, Dublin, Ireland

Lead weekly tutorials and support problem-solving sessions aligned with lectures.

Grade assignments and exams, providing clear and constructive feedback.

Jun 2024 – Dec 2025 Software Engineer (EdTech)

Grinds360, Dublin, Ireland

Built Python automation tools to streamline content and workflow pipelines for videos and interactive lessons.

Worked as part of product development teams, including UX iteration in Figma and full-stack feature implementation.

Collaborated with stakeholders to translate user needs into technical features, improving platform usability and reliability.

Jun 2025 – Sep 2025 Research Assistant (CRANN)

Trinity College Dublin (CRANN Institute), Dublin, Ireland

Designed and implemented ESN neural network reservoir computing models to learn and predict the time evolution of quantum systems from simulation data. Systematically analysed model stability, convergence, and predictive accuracy across a range of reservoir hyperparameters.

May 2024 – Sep 2024 Research Assistant (Experimental / Computational)

Trinity College Dublin, Dublin, Ireland

Created a control system for spectroscopy measurements on ferrofluids and liquid crystals under different experimental conditions.

Built Python tools for automated data acquisition and experiment control.

Jul 2021 – Jan 2024 Exam Corrector

The Dublin Academy of Education, Dublin, Ireland

Corrected exams and coursework; supported marking and feedback.

Projects / Research Highlights

Echo State Networks for Quantum Dynamics (2025)

Reservoir computing models for learning quantum time evolution; evaluated performance against numerical baselines.

Automated Spectroscopy Data Pipeline (2024)

Python-based tooling for automated experiment control and structured data logging. [arXiv:2504.19633](https://arxiv.org/abs/2504.19633)

Simulation and Data Analysis Tooling (2024–2025)

Developed modular Python and C/C++ codebases for numerical simulations, data processing, and performance analysis, with emphasis on correctness, efficiency, and reproducibility.